

#3. Basis for a Topology.

$$\#1. A = \bigcup_{x \in A} U_x.$$

#3. The Co-Countable is Topology, but τ_{co} is Not Topology.
[Counter example.

Let $X = \mathbb{R}$. Let $U_x = \mathbb{R} \setminus \{x\}$. Then, $U_x \in \tau_{co}$, but for fixed $x \in \mathbb{R}$, $y \in \mathbb{R}$,

$$\bigcup_{x \neq y} U_x = \mathbb{R} \setminus \{y\} \notin \tau_{co}.$$

That is, τ_{co} is Not closed under the union.

#4. a). Let $\{\tau_\alpha\}$ be a family of Topologies on X .

Then, $\bigcap \tau_\alpha$ is also Topology on X . But $\bigcup \tau_\alpha$ is not.

Proof. Clearly $\emptyset, X \in \bigcap \tau_\alpha$. And, let $U_i \in \bigcap \tau_\alpha$ ($i \in \mathbb{Z}$) be given

Then, $U_i \in \tau_\alpha$ for all $i \in \mathbb{Z}$ and $\alpha \in \Lambda$, then $\bigcup_{i \in \mathbb{Z}} U_i \in \tau_\alpha$ for all $\alpha \in \Lambda$.

So it is contained in $\bigcap \tau_\alpha$. Similarly, closed under the 'intersection'.

Counter example: $X = \{a, b, c\}$, $\tau_1 = \{\emptyset, X, \{a\}, \{a, b\}\}$, $\tau_2 = \{\emptyset, X, \{a\}, \{b, c\}\}$,

Then, $\{a, b\} \cap \{b, c\} = \{b\} \notin \tau_1 \cup \tau_2$

b). Let $S = \{U \mid U \in \tau_\alpha \text{ for some } \alpha\}$. Then, it is subbasis for X ,

and generates the smallest topology on X containing all τ_α .

The unique largest topology is $\bigcap \tau_\alpha$.

#5. If A is a basis of (X, τ) , then $\tau_A = \bigcap_{A \subseteq \tau'} \tau'$
Proof. Let $U \in \tau_A$ be given.
 τ' is topology.

Then, for some $\mathcal{O} \subseteq A$, $U = \bigcup \mathcal{O} \in \bigcap \tau'$.

The converse is trivial.

#6. Let $(-1,1) \setminus K \in \mathcal{U}_K$ be given. Then, $(-1,1) \setminus K \notin \mathcal{U}_I$, because for $0 \in (-1,1) \setminus K$, there is no basis element of $\mathcal{U}_I$ s.t. $0 \in B \subseteq (-1,1) \setminus K$. Conversely, $[0,1) \in \mathcal{U}_I$ is not contained in $\mathcal{U}_K$, because for $0 \in [0,1)$, there is no basis element of $\mathcal{U}_K$ s.t. $0 \in B \subseteq [0,1)$.

#7,

#8. b) The given $\mathcal{C}$ cannot generate $[\sqrt{2}, 3)$.

If $[\sqrt{2}, 3) \in \mathcal{U}_{\mathcal{C}}$, then for some $a_i \in \mathbb{Q}$ ($i \in \mathbb{Z}$),

$$[\sqrt{2}, 3) = \bigcup_{i \in \mathbb{Z}} [a_i, b_i)$$

Then, $\sqrt{2} \in [a_j, b_j)$ for some $j \in \mathbb{Z}$, and since $a_j \in \mathbb{Q}$, $a_j < \sqrt{2}$.

Contradiction.

Subspace.

#5. Let $(X, \mathcal{C}_1)$ and $(X, \mathcal{C}_2)$ be Topological spaces with $\mathcal{C}_1 \subseteq \mathcal{C}_2$ and let $(Y, \mathcal{C}_3)$ and $(Y, \mathcal{C}_4)$ be Topological spaces with $\mathcal{C}_3 \subseteq \mathcal{C}_4$. Then, $(X, \mathcal{C}_1) \times (Y, \mathcal{C}_3)$ is coarser than $(X, \mathcal{C}_2) \times (Y, \mathcal{C}_4)$.

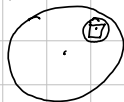
a) is trivial. To show the converse, let $U \in \mathcal{C}_2$ be given.

Then, $U \times Y \subseteq X \times Y$ is open, and $X' \times Y'$ is finer than $X \times Y$, $U \times Y \subseteq X' \times Y'$ is open.

Now, $\pi[U \times Y] = U \subseteq X'$ is open. $\square$

#6. Let $B_r(x) \subseteq \mathbb{R}^2$ be given. Then, for any $y \in B_r(x)$, put

$$\epsilon = r - d(x, y).$$



Finally, take $a, b, c, d \in \mathbb{Q}$ s.t.

$$y_1 < a < y_1 + \epsilon/\sqrt{2}, \quad y_1 - \epsilon/\sqrt{2} < b < y_1$$

$$y_2 < c < y_1 + \epsilon/\sqrt{2}, \quad y_2 - \epsilon/\sqrt{2} < d < y_2.$$

#7. Let $X = \mathbb{R} \setminus \{0\}$, $Y = (0, \infty)$.

Then Y is convex, but neither ray nor interval.
($0 \notin Y$)

#8. If L is a graph of $x = a$ for some $a \in \mathbb{R}$,

then $L \cong \mathbb{R}$ under the subspace of $\mathbb{R}_d \times \mathbb{R}$,

and $L \cong \mathbb{R}_d$ " $\mathbb{R}_d \times \mathbb{R}_d$.

And, in another case, $L \cong \mathbb{R}_d$.

#9.

Let $\mathcal{U} \subseteq \mathbb{R}^2$ be an basis of $\mathbb{R}^2$ with dic. order.

Let $(x, y) \in \mathcal{U}$ be given. Since $\mathbb{R}_d$ is discrete, $\{x\}$ is open of $\mathbb{R}_d$.

Meanwhile, the $\mathcal{U} = \{(a, b), (c, d) \mid (a, b) < (c, d)\}$, for this y ,

there is $\varepsilon > 0$ s.t. $(x, y - \varepsilon) < (x, y) < (x, y + \varepsilon)$ and $(x, y \pm \varepsilon) \in \mathcal{U}$.

Now, $(x, y) \in \{x\} \times (y - \varepsilon, y + \varepsilon) \subseteq \mathcal{U}$, thus $\mathbb{R}_d \times \mathbb{R}$ is finer than $\mathbb{R}^2$, dic.

Conversely, let $\{x\} \times (c, d) \subseteq \mathbb{R}_d \times \mathbb{R}$ be a basis element.

Then, it is clearly open of $\mathbb{R}^2$ dic. order.

#10. $\mathbb{I}^2 \neq \mathbb{I}_d^2$, because $(\frac{1}{2}) \times (\frac{1}{3}, \frac{2}{3})$ is open of $\mathbb{R}_d$, but not open in $\mathbb{R}^2$.

But, any basis of $\mathbb{I}^2$ is open of $\mathbb{I}_d^2$, because

$$(x_1, x_2) \in \mathbb{B}_r(x) \cap \mathbb{I}^2,$$

there is $\varepsilon > 0$ s.t. $(y_1, y_2) \in ((x_1, y_2 - \varepsilon), (x_1, y_2 + \varepsilon)) \cap \mathbb{I}^2 \subseteq \mathbb{B}_r(x) \cap \mathbb{I}^2$.

Section 17. Closed sets and Limit Points

#2. If A is closed in Y and Y is closed in X , then A is closed in X .

Proof. Y is closed in $X \Leftrightarrow X \setminus Y$ is open of X

A is closed in $Y \Leftrightarrow Y \setminus A$ is open of Y

$\Leftrightarrow Y \setminus A = Y \cap U$ for some open $U \subseteq X$

$X \setminus A = (X \setminus Y) \cup (Y \setminus A) = (X \setminus Y) \cup (Y \cap U) = (X \setminus Y) \cup U$ is open of X .

#3. If $A \subseteq X$ closed and $B \subseteq Y$ closed, then $A \times B \subseteq X \times Y$ is closed.

Proof. $(X \times Y) \setminus (A \times B) = (X \setminus A) \times Y \cup X \times (Y \setminus B)$

#4. Let Y be a Hausdorff space, X be any space, and $A \subseteq X$ be dense in X .

If $f, g: X \rightarrow Y$ are continuous and $f|_A = g|_A$, then $f = g$.

Proof. Let $x \in X$ be given. Suppose that $f(x) \neq g(x)$. Then, there are open $U, V \subseteq Y$

$f(x) \in U, g(x) \in V, U \cap V = \emptyset$.

Meanwhile, since f and g both are continuous, $f^{-1}[U]$ and $g^{-1}[V]$ are opens containing x .

Therefore, $f^{-1}[U] \cap g^{-1}[V] \cap A$ is non-empty open set, thus

$$f^{-1}[U] \cap g^{-1}[V] \cap A \neq \emptyset$$

since A is dense in X . Let $a \in f^{-1}[U] \cap g^{-1}[V] \cap A$. Then,

$f(a) \in U, g(a) \in V, a \in A$, thus $f(a) = g(a)$, $U \cap V \neq \emptyset$.

It is Contradiction.

#5. $(a,b) \subseteq [a,b]$ is trivial so enough to show that $(a,b)' \subseteq [a,b]$.

Let $\alpha \in X$ be a limit point of (a,b) . That is, for any open U containing α ,

$$(U \setminus \{\alpha\}) \cap (a,b) \neq \emptyset. \quad (*)$$

If $\alpha \notin [a,b]$, that is, $\alpha < a$ or $\alpha > b$. Either $\alpha < a$ or $\alpha > b$, see then:

$\alpha < a \Rightarrow [m, a)$ (if $m \in X$ is the smallest element) contains α , but does not satisfy $(*)$.

" $\Rightarrow (r, a)$ (if such m does not exist) $\neq$

And similarly in case of $\alpha > b$.

The equality cannot hold when $X = \mathbb{N}$, $(1, 3)$.

#6. a) is clear, by definition.

b) is shown by:

$$x \in \overline{A \cup B} \Leftrightarrow \text{for any open } U \text{ containing } x, U \cap (A \cup B) \neq \emptyset$$

and since $U \cap (A \cup B) = (U \cap A) \cup (U \cap B)$, either $U \cap A$ or $U \cap B$ is non-empty. So, $x \in \overline{A}$ or $x \in \overline{B}$. The converse is trivial by a).

c). For each B , $A_p \subseteq \bigcup A_\alpha$, so $\overline{A_p} \subseteq \overline{\bigcup A_\alpha}$.

But the converse is false, by the counterexample: $X = \mathbb{R}$, $A_n = \{\frac{1}{n}\}$. Then,

$$\bigcup_{n \in \mathbb{N}} \overline{A_n} = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\} \text{ but } \bigcup_{n \in \mathbb{N}} \overline{A_n} = \{\frac{1}{n} \mid n \in \mathbb{N}\}.$$

#8. a), $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$. Let $X = \mathbb{R}$, $A = (0, 1)$, $B = (1, 2)$. Then

$$\overline{A \cap B} = \overline{\emptyset} = \emptyset \text{ but } \overline{A} \cap \overline{B} = \{1\}.$$

And, since $\overline{A \cap B} \subseteq \overline{A}$, $\overline{B}$, so $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.

b) has the same result.

c). Let $X = \mathbb{R}$, $A = [0, 1)$, $B = (0, 1)$. Then

$$\overline{A - B} = \{0\} \text{ but } \overline{A} - \overline{B} = \emptyset. \text{ So } \overline{A - B} \subset \overline{A} - \overline{B} \text{ does not hold.}$$

And, let $x \in \overline{A - B}$ be given. That is, $x \in \overline{A}$ but $x \notin \overline{B}$.

Let U be an open containing x s.t. $U \cap B = \emptyset$. Since $x \in \overline{A}$, $U \cap A \neq \emptyset$.

So, $U \cap (A - B) = (U \cap A) \cap (U \cap B^c) = (U \cap A) \cap (U - B) = U \cap A \neq \emptyset$

Thus $\overline{A - B} \subseteq \overline{A - B}$.

#9.

$(x, y) \in \overline{A} \times \overline{B} \Leftrightarrow$ for any open $U \subseteq A$, $V \subseteq B$ containing x, y , respectively, $U \cap A \neq \emptyset$ and $V \cap B \neq \emptyset$.
 $\Leftrightarrow$ for any basis element $U \times V \subseteq X \times Y$ containing (x, y) , $(U \times V) \cap (A \times B) \neq \emptyset$
 $\Leftrightarrow (x, y) \in \overline{A \times B}$.

#11. Let X and Y be Hausdorff spaces.

Given distinct $(x_1, y_1), (x_2, y_2) \in X \times Y$, either $x_1 \neq x_2$ or $y_1 \neq y_2$ must be hold
 WLOG, suppose $y_1 \neq y_2$. So, there are opens $U_1, U_2 \subseteq Y$ s.t.

$$U_1 \cap U_2 = \emptyset, \quad y_1 \in U_1, \quad y_2 \in U_2.$$

So, $(X \times U_1) \cap (X \times U_2) = X \times (U_1 \cap U_2) = \emptyset$ and

$(x_i, y_i) \in X \times U_i \quad (i=1, 2)$. And these are clearly open, by definition.

#12. Let X be a Hausdorff and $A \subseteq X$.

Given two distinct $x, y \in A$, there are opens $U, V \subseteq X$ s.t. $x \in U$, $y \in V$, $U \cap V = \emptyset$.

Now, $(U \cap A) \cap (V \cap A) = \emptyset$, containing x, y .

#13. X is $T_2 \Leftrightarrow \forall (x_1, x_2) \in X \times X, (x_1, x_2) \notin \Delta \Rightarrow \exists \underbrace{U, V}_{\text{open}} \subseteq X$ s.t. $(x_1, x_2) \in U \times V$
 $U \cap V = \emptyset \Leftrightarrow (U \times V) \cap \Delta = \emptyset$
 $\Leftrightarrow U \times V \subseteq \Delta^c$

$\Leftrightarrow \Delta$ is closed.

#14. $x = \lim x_n \Leftrightarrow$ Given open $U \subseteq \mathbb{R}$ containing x , there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow x_n \in U$

Since for any open $U \subseteq \mathbb{R}$, $\mathbb{R} \setminus U$ is finite, put $\# \mathbb{R} \setminus U = N$. Then,

x_n must be contained in U , for all $n \geq N$

So, the answer is $\mathbb{R}$.

#19. Def: $\text{Bd } A = \overline{A} \cap \overline{X-A}$.

a). $\text{Int } A \cap \text{Bd } A = \emptyset$.

Proof. Suppose that $\text{Int } A \cap \text{Bd } A \neq \emptyset$. Let $x \in \text{Int } A \cap \text{Bd } A$.

Then, there is an open U containing x s.t. $x \in U \subseteq A$.

And, since $x \in \text{Bd } A = \overline{A} \cap \overline{X-A}$,

$$U \cap A \neq \emptyset \quad \text{and} \quad U \cap (X-A) \neq \emptyset$$

But $U \cap (X-A) \neq \emptyset \Leftrightarrow U \not\subseteq A$, so it is contradiction.

And, since $\text{Int } A, \text{Bd } A \subseteq \overline{A}$, $\text{Int } A \cup \text{Bd } A \subseteq \overline{A}$.

Conversely, for $x \in \overline{A}$, suppose that $x \notin \text{Int } A$. Then, there is an open U containing x s.t.

$$x \in U \quad \text{but} \quad U \not\subseteq A$$

$$U \not\subseteq A \Leftrightarrow U \cap (X-A) \neq \emptyset, \text{ so } x \in \text{Bd } A. \quad \square$$

b). $\text{Bd } A = \emptyset \Leftrightarrow \overline{A} = \text{Int } A$.

If A is clopen, $\overline{A} = A = \text{Int } A$. And, if $\overline{A} = \text{Int } A$, then

$$A \subseteq \overline{A} = \text{Int } A, \text{ so } A = \text{Int } A, \text{ and } \overline{A} = \text{Int } A \subseteq A, \text{ so } \overline{A} = A$$

Section 18.

#2. Let $V \subset Y$ be an open containing $f(x)$. Then,

$f^{-1}[V]$ is open of X , containing x .

Since x is the limit point of A , $f^{-1}[V] \cap A \neq \emptyset$. Now,

$$V \cap f[A] \supseteq f[f^{-1}[V] \cap f[A]] = f[f^{-1}[V] \cap A] \neq \emptyset,$$

so $f(x)$ is the limit point of $f[A]$.

#6. $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$

f is continuous at only $x=0$.

#10. Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be continuous maps. Then, a map

$$(f, g): A \times C \rightarrow B \times D; (a, c) \mapsto (f(a), g(c))$$

is continuous.

Proof. Given a basis element $U \times V \subseteq B \times D$, $f^{-1}[\pi_A[U \times V]]$ and $g^{-1}[\pi_B[U \times V]]$

$$(x, y) \in (f, g)^{-1}[U \times V] \Leftrightarrow (f(x), g(y)) \in U \times V \Leftrightarrow f(x) \in U \text{ and } g(y) \in V \Leftrightarrow (x, y) \in f^{-1}[U] \times g^{-1}[V]$$

So, given $(x, y) \in (f, g)^{-1}[U \times V]$, $f(x) \in U$ and $g(y) \in V$.

Since f and g are continuous, there are open $M \subseteq A$ and open $N \subseteq C$ such that

$$x \in M \text{ and } f[M] \subseteq U, \quad y \in N \text{ and } g[N] \subseteq V.$$

Therefore, $(x, y) \in M \times N \subseteq f^{-1}[U] \times g^{-1}[V] = (f, g)^{-1}[U \times V]$

#11. Let $F: X \times Y \rightarrow Z$ be a map.

We say that F is continuous in each variable separately if:
for any $y_0 \in Y$, $F_x: X \rightarrow Z: x \mapsto F(x, y_0)$ is continuous, and
for any $x_0 \in X$, $F_y: Y \rightarrow Z: y \mapsto F(x_0, y)$ is continuous.

Show that F is continuous $\Rightarrow F$ is continuous in each variable separately

But the converse does not hold.

Proof. Let $y_0 \in Y$ be fixed. Then, a map

$$e_{y_0}: X \rightarrow X \times Y: x \mapsto (x, y_0)$$

is embedding. That is, continuous, open, one-to-one map.

Continuous: for any basis $U \times V \subseteq X \times Y$, $e_{y_0}^{-1}[U \times V] = \begin{cases} U & \text{if } y_0 \in V \\ \emptyset & \text{if } y_0 \notin V \end{cases}$

either $y_0 \in V$ or not, the preimage preserves open

open: for any open $U \subseteq X$, $U \times \{y_0\}$ is the open set of the subspace $X \times \{y_0\}$.

Now, $F_x = e_{y_0} \circ F$ is composition of continuous maps, so open.

Similarly F_y .

But, there is a counterexample for the converse:

$$F: \mathbb{R}^2 \rightarrow \mathbb{R}: (x, y) \mapsto \begin{cases} 0 & \text{if } (x, y) = (0, 0) \\ \frac{xy}{x^2 + y^2} & \text{otherwise} \end{cases}$$

Section 1a.

+ . $f: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}; t \mapsto (t, t, \dots, t)$ is continuous under the product topology, but not continuous under the box topology.

Proof. Let $A = \prod_{n=1}^{\infty} (-\frac{1}{n}, \frac{1}{n})$ be given. Clearly, A is open of $\mathbb{R}^{\mathbb{N}}$ with box topology. But, $f^{-1}[A] = \{0\}$, which is not open of $\mathbb{R}$, so f is not continuous.

#6. Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence in $\prod_{\alpha \in \Lambda} X_\alpha$. Then,

A_n converges to A if and only if for each α , $\{\pi_\alpha(A_n)\}_{n \in \mathbb{N}}$ converges to $\pi_\alpha(A)$.

Proof. A_n converges to A if and only if

for any basis element $B \subseteq \prod_{\alpha \in \Lambda} X_\alpha$ containing A , there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$

Suppose that $A_n \rightarrow A$. Given $\alpha \in \Lambda$, let $U_\alpha \subseteq X_\alpha$ be an open containing $\pi_\alpha(A)$.

Then, $B = \prod_{\alpha \in \Lambda} U_\alpha$ where $U_\beta = X_\beta$ if $\beta \neq \alpha$ is basis element containing A , so

there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$. That is, $\pi_\alpha(A_n) \in U_\alpha = \pi_\alpha[B]$, $\forall n \geq N$.

Conversely, let $B \subseteq \prod_{\alpha \in \Lambda} X_\alpha$ be a basis element containing $A \in B$.

By definition, $B = \prod_{\alpha \in \Lambda} U_\alpha$ s.t. all $U_\alpha = X_\alpha$ but finitely many $\beta \in \Lambda$,

let such $\beta \in \Lambda$ as $\beta_1, \dots, \beta_k$. By the assumption, for each β_i , there is $N_i \in \mathbb{N}$ s.t. $n \geq N_i \Rightarrow \pi_{\beta_i}(A_n) \in U_{\beta_i}$.

Take $N = \max\{N_1, \dots, N_k\}$. Now,

$$n \geq N \Rightarrow A_n \in B. \quad \square$$

In box topology, $\Rightarrow$ holds but does not $\Leftarrow$.

#7. $\mathbb{R}^\infty = \bigcup_{n=1}^{\infty} \prod_{k=1}^n \mathbb{R}_k^n$ where $\mathbb{R}_k^n = \begin{cases} \{0\} & k > n \\ \mathbb{R} & k \leq n \end{cases}$

$A \in \mathbb{R}^{\mathbb{N}}$ is contained in $\overline{\mathbb{R}^\infty}$ if and only if:

for any basis $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ containing A , $\mathcal{U} \cap \mathbb{R}^\infty \neq \emptyset$.

Under the Product Topology, for given $A \in \mathbb{R}^{\mathbb{N}}$, denote $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ as

$$\mathcal{U} = \bigcap_{n=1}^{\infty} \mathcal{U}_n \quad (\mathcal{U}_n = \{B \mid \text{all } n \text{ neighbors but finitely many } n_i \text{ and } \pi_n(A) \in \mathcal{U}_n, \forall n \in \mathbb{N}\})$$

$$\begin{aligned} \text{Then, } \mathcal{U} \cap \mathbb{R}^\infty &= \left(\bigcap_{n=1}^{\infty} \mathcal{U}_n \right) \cap \left(\bigcup_{n=1}^{\infty} \left(\left[\prod_{k=1}^n \mathbb{R} \right] \times \left[\prod_{k=n+1}^{\infty} \{0\} \right] \right) \right) \\ &= \bigcup_{n=1}^{\infty} \left[\left(\bigcap_{k=1}^{\infty} \mathcal{U}_k \right) \cap \left(\prod_{k=1}^n \mathbb{R}_k^n \right) \right] \text{ where } \mathbb{R}_k^n = \begin{cases} \{0\} & k > n \\ \mathbb{R} & k \leq n \end{cases} \\ &= \bigcup_{n=1}^{\infty} \prod_{k=1}^{\infty} (\mathcal{U}_k \cap \mathbb{R}_k^n) \end{aligned}$$

It cannot be empty set because, if $m \in \mathbb{N}$ is a number s.t. $k \geq m \Rightarrow \mathcal{U}_k = \mathbb{R}$, then

$$\prod_{k=1}^{\infty} \mathcal{U}_k \cap \prod_{k=1}^{\infty} \mathbb{R}_k^m \neq \emptyset$$

Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^{\mathbb{N}}$. But, under the box topology, it cannot be held, for

$$\mathcal{U} = \prod_{n=1}^{\infty} (0,1)$$

valid as counterexample. Actually, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$. (Becomes no such $A \in \overline{\mathbb{R}^\infty}$ with $A \notin \mathbb{R}^\infty$.)

#8. h is bijective, with the inverse as

$$h^{-1}: \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}: (x_n)_{n \in \mathbb{N}} \mapsto \left(\frac{1}{a_n} \cdot [x - b_n] \right)$$

And, regardless of box or prod. topol.

#10. Subbasis with Initial Topology.

Let X be a set, and $\{X_\alpha\}_{\alpha \in A}$ be a collection of spaces.

1. Given $f_\alpha: X \rightarrow X_\alpha$ ($\alpha \in A$), there exists a unique coarsest Topology on X s.t.
for all $\alpha \in A$, f_α is continuous map.

Proof. Let $S_\alpha \stackrel{\text{def}}{=} \{f_\alpha^{-1}[U_\alpha] \mid U_\alpha \subseteq X_\alpha \text{ is open of } X_\alpha\}$

Then, $S \stackrel{\text{def}}{=} \bigcup_{\alpha \in A} S_\alpha$ is a subbasis of X , because: $X = f_\alpha^{-1}[X_\alpha]$, for any $\alpha \in A$.

Given $\alpha \in A$, and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha] \in S_\alpha \subseteq S$, so all f_α are continuous.

Moreover, it is the Coarsest Topology s.t. all f_α are continuous, because:

if $\mathcal{T}$ is a Topology on X satisfying such condition,

then for any $\alpha \in A$ and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha]$ is contained in $\mathcal{T}$,
therefore $\mathcal{T}_S \subseteq \mathcal{T}$.

Note: Such a Topology on X is called the initial Topology for $\{f_\alpha\}$.

2. A map $g: Y \rightarrow X$ is continuous iff $f_\alpha \circ g$ is continuous for all $\alpha \in A$.

(The given Y is any space, and X is the initial Topology.)

Proof. Since the Composition of Continuous maps is Continuous $\Rightarrow$ is trivial.

Conversely, By the assumption, for each $\alpha \in A$, and an open $U_\alpha \subseteq X_\alpha$

$(f_\alpha \circ g)^{-1}[U_\alpha] = g^{-1}[f_\alpha^{-1}[U_\alpha]]$ is an open of Y .

Meanwhile, by the first assertion, every open of X is generated by S ,

$g^{-1}[U]$ is open for any open $U \subseteq X$.

Section Metric.

#2. Define $d: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R} : (x_1, y_1), (x_2, y_2) \mapsto \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ 1 + |y_1 - y_2| & \text{if } x_1 \neq x_2. \end{cases}$

Then, for any open set U of $\mathbb{R} \times \mathbb{R}$ with the dictionary order topology, let $(x, y) \in U$ be given. Then, there are $(x_1, y_1), (x_2, y_2) \in U$ s.t.

$$(x_1, y_1) < (x, y) < (x_2, y_2)$$

Take $r = \min\{|y_1 - y|, |y_2 - y|\}$. Then, $B_r(x, y) \subseteq U$.

Conversely, let $B_r(x, y)$ be given as arbitrary open ball.

Take $y_1 = y - \frac{r}{2}$ and $y_2 = y + \frac{r}{2}$. Then,

$$((x, y_1), (x, y_2)) \subseteq B_r(x, y) \quad \square$$

#3. Let (X, d) be a Metric Space. Then, the Metric Topology generated by

$$\beta_d := \{B_r(x) \mid r > 0, x \in X\}$$

is the Coarsest Topology such that $d: X \times X \rightarrow \mathbb{R}$ is Continuous.

Proof. Given a basis element $(a, b) \subset \mathbb{R}$ ($a < b$),

$$d^{-1}[(a, b)] = \{(x, y) \in X \times X \mid d(x, y) \in (a, b)\}.$$

If $b \leq 0$, then the preimage above is empty, so ok. So, assume that $b > 0$.

Given $(x, y) \in d^{-1}[(a, b)]$, $a < d(x, y) < b$. Take $r_x, r_y > 0$ such that

$$r_x + r_y < \min\{d(x, y) - a, b - d(x, y)\}.$$

Now,

$$(x, y) \in B_{r_x}(x) \times B_{r_y}(y) \subseteq d^{-1}[(a, b)], \text{ because for any } (x', y') \in B_{r_x}(x) \times B_{r_y}(y),$$

$$d(x', x) < r_x \text{ and } d(y, y') < r_y, \text{ so}$$

$$d(x', y') \leq d(x', x) + d(x, y) + d(y, y') < r_x + d(x, y) + r_y < b$$

$$\text{and } d(x, y) \leq d(x, x') + d(x', y') + d(y', y) < r_x + d(x', y') + r_y, \text{ so}$$

$$a < d(x, y) - (r_x + r_y) < d(x', y').$$

$$\text{Thus is, } d(x', y') \in (a, b), \text{ or } (x', y') \in d^{-1}[(a, b)].$$

To show that the second assertion, let $\mathcal{T}_d$ be a Topology on X such that

$$d: X \times X \rightarrow \mathbb{R}$$

is Continuous. Since every Continuous map on the Product Topology is Continuous each variable, given $x_0 \in X$, $f_{x_0}: X \rightarrow \mathbb{R}; x \mapsto d(x, x_0)$ is Continuous on $(X, \mathcal{T}_d)$.

We want to show that $\mathcal{T}_d \subseteq \beta_d$, so, let $B_r(x_0) \in \mathcal{T}_d$ be given.

$$\text{for any } y \in B_r(x_0), B_\epsilon(y) \subseteq B_r(x_0), \text{ where } \epsilon = r - d(x_0, y).$$

$$\text{Meanwhile, } B_\epsilon(y) = f_y^{-1}[-\infty, \epsilon], \text{ so } B_r(x_0) \text{ is open at } (X, \mathcal{T}_d).$$

Thus Proved.

#4.

$$f: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}; t \mapsto (t, 2t, 3t, \dots)$$

$$g: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}; t \mapsto (t, t, \dots)$$

$$h: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}; t \mapsto (t, \frac{1}{2}t, \frac{1}{3}t, \dots)$$

T.P	Continuous table		
	Product	box	Uniform
f	0	X	X
g	0	X	0
h	0	X	0

$(1,1)$: the preimage is finite intersection of opens.

$(2,1)$: "

$(3,1)$: "

But, in the box topology, let $U = \prod_{n=1}^{\infty} (-\frac{1}{n^2}, \frac{1}{n^2})$. Then,

$$nt \in (-\frac{1}{n^2}, \frac{1}{n^2}) \Leftrightarrow t \in (-\frac{1}{n^3}, \frac{1}{n^3}) \text{ and}$$

$$t \in (-\frac{1}{n^2}, \frac{1}{n^2})$$

$$\frac{1}{n}t \in (-\frac{1}{n^4}, \frac{1}{n^4}) \Leftrightarrow t \in (-\frac{1}{n^3}, \frac{1}{n^3})$$

So, each f, g, h , the preimages are $\{0\}$, closed. Thus Not Continuous.

Recall: (X, d) , $\bar{d}: X \times X \rightarrow \mathbb{R}$ $\bar{d}(x, y) = \min\{d(x, y), 1\}$

$$(\mathbb{R}^{\mathbb{N}}, d). \rho: \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}; (\{x_n\}, \{y_n\}) \mapsto \sup\{\bar{d}(x_n, y_n) \mid n \in \mathbb{N}\}.$$

Given $x \in \mathbb{R}$ and $\epsilon > 0$, take $\delta = \frac{\epsilon}{2}$

Then, $|x - y| < \delta \Rightarrow \rho(f(x), f(y)) = \sup_{n \in \mathbb{N}} \min\{|nx - ny|, 1\} \quad (\text{false})$

$$|x - y| < \delta \Rightarrow \rho(g(x), g(y)) = \sup_{n \in \mathbb{N}} \min\{|x - y|, 1\} < \epsilon$$

$$|x - y| < \delta \Rightarrow \rho(h(x), h(y)) = \sup_{n \in \mathbb{N}} \min\{\frac{1}{n}|x - y|, 1\} < \epsilon$$

□

Sec 20.5.

$$\text{Let } \mathbb{R}^\infty = \bigcup_{n=1}^{\infty} \underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_n \times \underbrace{\{0\} \times \{0\} \times \dots}_{\infty} \subseteq \mathbb{R}^{\mathbb{N}}.$$

What is $\overline{\mathbb{R}^\infty}$ under the Uniform Topology?

Proof. $\mathbb{R}^\infty \subseteq \overline{\mathbb{R}^\infty}$ is clear, give the property of closure.

Let $\{x_n\} \in \mathbb{R}^{\mathbb{N}}$ be a sequence s.t. $x_n \rightarrow 0$ as $n \rightarrow \infty$.

Then, given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |x_n| < \varepsilon$ or $0 \in B_\varepsilon(x_n)$.

That is, given $\varepsilon > 0$, $\mathbb{R}^\infty \cap B_{2\varepsilon}(\{x_n\})$ is not empty,

for fixed $\underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_N \times \{0\} \times \dots$, $x_1, \dots, x_{N-1} \in \mathbb{R}$, and $|x_k| < \varepsilon$ for all $k \geq N$, so

$$(x_1, x_2, \dots, x_{N-1}, 0, 0, \dots) \in \mathbb{R}^\infty \cap B_{2\varepsilon}(\{x_n\}).$$

So, $\overline{\mathbb{R}^\infty} = \{x_n \mid x_n \rightarrow 0 \text{ as } n \rightarrow \infty\}$,

(If $\{x_n\} \not\rightarrow 0$, then we can take $\varepsilon > 0$ s.t. $\mathbb{R}^\infty \cap B_\varepsilon(\{x_n\}) = \emptyset$.)

More rigorously, there is $\varepsilon > 0$ s.t. for any $N \in \mathbb{N}$, there is $n \geq N$ s.t. $|x_n| \geq \varepsilon$.

So, for any $\mathbb{R}^{\mathbb{N}} \times \{0\}^\infty$, $(\mathbb{R}^{\mathbb{N}} \times \{0\}^\infty) \cap B_\varepsilon(\{x_n\}) = \emptyset$, so $\mathbb{R}^\infty \cap B_\varepsilon(\{x_n\}) = \bigcup \emptyset = \emptyset$.

20, 6.

Let $\bar{p}$ be the uniform metric on $\mathbb{R}^N$.

Given $(x_n) \in \mathbb{R}^N$, and $0 < \varepsilon < 1$, define

$$\mathcal{U}((x_n), \varepsilon) \stackrel{\text{def}}{=} \prod_{n=1}^{\infty} (x_n - \varepsilon, x_n + \varepsilon)$$

a). $\mathcal{U}((x_n), \varepsilon) \neq B_{\bar{p}}((x_n), \varepsilon)$. $\bar{p}: \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}; ((x_n), (y_n)) \mapsto \sup_{n \in \mathbb{N}} \min\{|x_n - y_n|, 1\}$.

b). $\mathcal{U}((x_n), \varepsilon)$ is Not open.

c). $B_{\bar{p}}((x_n), \varepsilon) = \bigcup_{\delta < \varepsilon} \mathcal{U}((x_n), \delta)$.

Proof. Let $y_n = x_n + \varepsilon \cdot \frac{n-1}{n}$. Then, for all $n \in \mathbb{N}$, $|x_n - y_n| = \varepsilon \cdot \frac{n-1}{n} < \varepsilon$, so $(y_n) \in \mathcal{U}((x_n), \varepsilon)$. but! $\bar{p}((x_n), (y_n)) = \sup_{n \in \mathbb{N}} \varepsilon \cdot \frac{n-1}{n} = \varepsilon$, so $(y_n) \notin B_{\bar{p}}((x_n), \varepsilon)$.

As seen above, $(y_n) \in \mathcal{U}((x_n), \varepsilon)$. Suppose that $\mathcal{U}((x_n), \varepsilon)$ is open. Then, there is $\delta > 0$ s.t. $(y_n) \in B_{\bar{p}}((y_n), \delta) \subseteq \mathcal{U}((x_n), \varepsilon)$.

Take $y'_n = y_n + \frac{\delta}{2}$. Then, $(y'_n) \in B_{\bar{p}}((y_n), \delta)$ but

$$y'_n = y_n + \frac{\delta}{2} = x_n + \frac{n-1}{n} \cdot \varepsilon + \frac{\delta}{2} \notin (x_n - \varepsilon, x_n + \varepsilon)$$

for some enough large $n \in \mathbb{N}$, so contradiction.

(specifically, take $n \in \mathbb{N}$ s.t. $\frac{n-1}{n} \cdot \varepsilon + \frac{\delta}{2} \geq \varepsilon$, or $\frac{\delta}{2} \geq \frac{\varepsilon}{n}$).

Given $\delta < \varepsilon$, $\mathcal{U}((x_n), \delta) \subseteq B_{\bar{p}}((x_n), \varepsilon)$ is trivial, so $B_{\bar{p}}((x_n), \varepsilon) \supseteq \bigcup_{\delta < \varepsilon} \mathcal{U}((x_n), \delta)$.

Now, let $(y_n) \in B_{\bar{p}}((x_n), \varepsilon)$ be given. Then, $\sup_{n \in \mathbb{N}} |x_n - y_n| < \varepsilon$, so for some $\delta < \varepsilon$, for all $n \in \mathbb{N}$, $|x_n - y_n| \leq \sup_{n \in \mathbb{N}} |x_n - y_n| < \delta < \varepsilon$. So proved. $\square$

#20, 8. ℓ^2 -space.

Define a metric on $X \stackrel{\text{def}}{=} \{(x_n) \in \mathbb{R}^{\mathbb{N}} \mid \sum_{n=1}^{\infty} x_n^2 < \infty\}$ as

$$d: X \times X \rightarrow \mathbb{R}; (x_n), (y_n) \mapsto \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2}.$$

The Topological Space (X, d) is called ℓ^2 -space.

a). Show that on X ,

$$\text{box topology} \supset \ell^2\text{-topology} \supset \text{uniform topology} \supset \text{Product topology}$$

Prod.

Sec 21.

#21.1. Let $A \subset X$. If $d: X \times X \rightarrow \mathbb{R}$ is a metric for the topology of X , show that $d|_{A \times A}: A \times A \rightarrow \mathbb{R}; (a, b) \mapsto d(a, b)$ is a metric for the subspace topology on A .

Proof. Let $(a, b) \in \mathbb{R}$ be given. Then,

$$d|_{A \times A}^{-1}([a, b]) = d|^{-1}([a, b]) \cap A \times A$$

So, by definition of subspace, it is continuous.

#21.2. Let (X, d_X) and (Y, d_Y) be metric spaces.

Let $f: X \rightarrow Y$ be a map satisfying

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2) \quad \text{for all } x_1, x_2 \in X.$$

Then, f is an imbedding.

Proof. 1). f is continuous

Let $x \in X$ be fixed. Given $\varepsilon > 0$, let $\delta = \varepsilon$. Then,

$$d_X(x, t) < \delta \Rightarrow d_Y(f(x), f(t)) = d_X(x, t) < \delta = \varepsilon, \text{ so it is continuous.}$$

2). f is one-to-one.

$$f(x_1) = f(x_2) \Leftrightarrow d_Y(f(x_1), f(x_2)) = 0 \Leftrightarrow d_X(x_1, x_2) = 0 \Leftrightarrow x_1 = x_2.$$

3). f is open map.

Let $B_r(x) \subseteq X$ be given. Then, $f[B_r(x)]$ is open in $f[X]$, because given $y \in f[B_r(x)]$, $y = f(t)$ for some $t \in B_r(x)$. Put $\varepsilon = r - d_X(x, t)$.

$$\text{Then, } y = f(t) \subseteq \underbrace{B_\varepsilon(f(t)) \cap f[X]}_{(*)} \subseteq f[B_r(x)]$$

(*) is given by $u \in B_\varepsilon(f(t)) \cap f[X]$, $u = f(s)$ for some $s \in X$ and $d_Y(f(t), f(s)) < \varepsilon$.

Then, $d_X(t, s) < \varepsilon = r - d_X(x, t)$, so $d_X(x, s) < r$.

This gives that $u = f(s) \in f[B_r(x)]$.

+ Additional

Let (X, d) be a Compact Metric space.

Suppose that $f: X \rightarrow X$ is a Continuous isometry.

Then, f is Homeomorphism.

Proof. 1). f is one-to-one

$$f(x_1) = f(x_2) \Rightarrow d(f(x_1), f(x_2)) = 0 \Rightarrow d(x_1, x_2) = 0 \Rightarrow x_1 = x_2.$$

Topological Group.

Def. Let G be a Group under the binary operation $\cdot: G \times G \rightarrow G$.

A Topological Space $(G, \mathcal{L})$ satisfying T_1 condition, and

$G \times G \rightarrow G: (x, y) \mapsto xy$ and $G \rightarrow G: x \mapsto x^{-1}$ are continuous

is called the Topological Group.

Lemma. Let G be a group and T_1 -space. Then

G is Topological Group if and only if $G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Proof. Suppose that G is Topological Group. Then

$\varphi(x, y) = xy$ and $\psi(y) = y^{-1}$ are continuous, therefore

$(x, y) \mapsto xy^{-1} = \varphi(x, \psi(y))$ is continuous, being composition of continuous maps

Conversely, suppose that $\mu: G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Then, for fixed $x = 1$, $\mu(1, y) = y^{-1}$ is continuous, so $x \mapsto x^{-1}$ is continuous.

Now, $y \mapsto y^{-1}$ is continuous, so

$(x, y) \mapsto x(y^{-1})^{-1} = xy$ is continuous, being composition of continuous.

□

Examples.

$(\mathbb{Z}, +)$ is clearly a Topological Group, because the Topology is discrete.

$(\mathbb{R}, +)$ is Topological Group under the Standard Topology.

Proof. T_1 is clear, by Metric $\Rightarrow T_1$

Let $\epsilon > 0$ be given. Then for $\delta = \epsilon$,

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} < \delta \Rightarrow |x_1 - y_1 - (x_2 - y_2)| = |(x_1 - x_2) - (y_1 - y_2)| \leq |x_1 - x_2| + |y_1 - y_2| < 2\epsilon.$$

Thus, it is a Topological Group.

$(\mathbb{R}_+, \cdot)$ is Topological Group.

Proof. Given $\epsilon > 0$,

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} < \delta \Rightarrow \left| \frac{x_1}{y_1} - \frac{x_2}{y_2} \right| = \frac{1}{|y_1 y_2|} \cdot |x_1 y_2 - x_2 y_1|$$

$GL_n(\mathbb{R})$.

Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$, with Euclidean space. Define a metric for $M_{nn}(\mathbb{R})$ as:

$$d: M_{nn}(\mathbb{R}) \times M_{nn}(\mathbb{R}) \rightarrow \mathbb{R}; (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu: M_{nn}(\mathbb{R}) \times M_{nn}(\mathbb{R}) \rightarrow M_{nn}(\mathbb{R}); (A, B) \mapsto \left(\sum_{k=1}^n A_{ik} \cdot B_{kj} \right)_{i,j}$$

is continuous, because it is entry-wise continuous.

Since $GL_n(\mathbb{R})$ is closed under the multiplication, it has also continuous operation.

And, $GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R}); A \mapsto A^{-1}$

is also continuous, because $A^{-1} = (\det A)^{-1} \cdot \text{adj } A$

$$\text{where } (\text{adj } A)_{i,j} = (-1)^{i+j} \cdot \det A(\hat{j} \hat{i}).$$

is also multivariables polynomial, entrywise.

Consider $O_n(\mathbb{R}) = \{A \in M_{nn}(\mathbb{R}) \mid A^t = -A\}$, $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$.

Then, O_n and SO_n are Compact, and SO_n is connected.

Proof. $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^n A_{ik} \cdot A_{jk} = \delta_{ij}$.

Define a map $f_{i,j}: M_{nn}(\mathbb{R}) \rightarrow \mathbb{R}; A \mapsto \sum_{k=1}^n A_{ik} \cdot A_{jk}$.

Then, it is continuous therefore

$$O_n(\mathbb{R}) = \bigcap_{\substack{1 \leq i, j \leq n \\ i \neq j}} f_{i,j}^{-1}[\{0\}] \cap \bigcap_{1 \leq i \leq n} f_{i,i}^{-1}[\{1\}] \text{ is finite intersection of closed sets,}$$

thus closed. Moreover, $\dim O_n(\mathbb{R}) \leq n$, so it is bounded. Therefore Compact.

And, $\det: O_n(\mathbb{R}) \rightarrow \mathbb{R}; A \mapsto \det A$ is continuous, so,

$SO_n(\mathbb{R}) = \det^{-1}[\{1\}]$ also closed, and bounded by subset of a bounded set.

Section 7, Connected, Start! Yeah!

1. Let $\mathcal{C}$ and $\mathcal{C}'$ be two topologies on X .

Suppose that $\mathcal{C} \subseteq \mathcal{C}'$. Answer:

a). If $(X, \mathcal{C}')$ is Connected, then $(X, \mathcal{C})$ is Connected?

b). Converse of a) holds?

Proof. Suppose that $(X, \mathcal{C})$ is disconnected. Let $\{U, V\}$ be the separation.

That is, $U \neq \emptyset$, $V \neq \emptyset$, $U \cap V = \emptyset$, $U \cup V = X$ and $U, V \in \mathcal{C}$.

Then, since $\mathcal{C} \subseteq \mathcal{C}'$, $\{U, V\}$ is also separation for $(X, \mathcal{C}')$.

Therefore, $(X, \mathcal{C})$ is disconnected $\Rightarrow (X, \mathcal{C}')$ is disconnected,

or If the bigger topology is Connected, then the contained topology also connected.

But, the Converse fails: $\mathbb{R}$, euclidean is Connected, but $\mathbb{R}$, equipped indiscrete is disconnected.

2. Let $\{A_n\}$ be a sequence of Connected Subspaces of X satisfying $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$.

Then $\bigcup A_n$ is Connected.

Proof. First, show that for $B_n = A_1 \cup \dots \cup A_n$, B_n is connected for all $n \in \mathbb{N}$.

For $n=1$, $B_1 = A_1$ is Connected. Suppose B_n is Connected, then, $B_{n+1} = B_n \cup A_{n+1}$ is Connected by

$$B_n \cap A_{n+1} = (A_1 \cup \dots \cup A_n) \cap A_{n+1} = (A_1 \cap A_{n+1}) \cup \dots \cup (A_n \cap A_{n+1}) \neq \emptyset, \text{ by the hypothesis}$$

So, by the theorem, B_{n+1} is Connected. Now,

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n \text{ is Connected, because } \bigcap_{n=1}^{\infty} B_n = A_1 \neq \emptyset \text{ since } A_1 \cap A_2 \neq \emptyset.$$

Lemmr. Let $A_\alpha \subset X$ ($\alpha \in I$) be connected subspaces and $\bigcap_{\alpha \in I} A_\alpha \neq \emptyset$, then $\bigcup_{\alpha \in I} A_\alpha$ is connected.

Proof. Using Contradiction: Suppose that $A = \bigcup_{\alpha \in I} A_\alpha$ is disconnected. Then, there are open of A , U, V , satisfying:

$$U \neq \emptyset, V \neq \emptyset, U \cap V = \emptyset, U \cup V = A.$$

Let $\alpha \in \bigcap_{\alpha \in I} A_\alpha$ be fixed. Then, $\alpha \in A = U \cup V$. WLOG, assume $\alpha \in U$. For each $\alpha \in I$, define $U_\alpha = U \cap A_\alpha$ and $V_\alpha = V \cap A_\alpha$. Then, for all $\alpha \in I$, $\alpha \in U_\alpha \neq \emptyset$, $U_\alpha \cap V_\alpha = \emptyset$, $U_\alpha \cup V_\alpha = (U \cup V) \cap A_\alpha = A_\alpha$.

And, set-theoretically U_α, V_α are open of A_α .

Since each A_α ($\alpha \in I$) is connected, $\{U_\alpha, V_\alpha\}$ cannot be the separation of A_α , so $V_\alpha = \emptyset$ for all $\alpha \in I$. But, this implies

$$V = V \cap A = V \cap \left(\bigcup_{\alpha \in I} A_\alpha \right) = \bigcup_{\alpha \in I} (V \cap A_\alpha) = \bigcup_{\alpha \in I} V_\alpha = \emptyset.$$

3. Let $\{A_\alpha\}$ be a collection of connected subspaces of X , let A be a connected subspace. If $A \cap A_\alpha \neq \emptyset$ for all $\alpha \in I$, then $A \cup \left(\bigcup_{\alpha \in I} A_\alpha \right)$ is connected.

Proof. Since $A \cap A_\alpha \neq \emptyset$ and both are connected, $A \cup A_\alpha$ is connected, by the Lemma above.

And, $\bigcap_{\alpha \in I} (A \cup A_\alpha) = A \cup \left(\bigcap_{\alpha \in I} A_\alpha \right) \neq \emptyset$, so it is connected.

#4. If X is infinite set equipped the finite complement topology, then X is connected.

Proof. If $\{U, V\} \in \tau_X$ is the separation, but, $U \cap V = \emptyset$ cannot possible, because $U \cap V = \emptyset$ equals to $U \subseteq V^c$ set-theoretically, but

U is clearly infinite set and V^c is finite, it is contradiction.

#5. The lower-limit Topology $\mathbb{R}_l$ is totally this connected.

+7 Every subset containing more than two points must be separated by the disjoint open
 [a) $[a, b)$, a is any point of the given subset, with a is not minimal element of S .

#6. Let $A \subseteq X$. If C is a connected subspace of X then

$$C \cap A \neq \emptyset \text{ and } C \cap (X \setminus A) \neq \emptyset.$$

then $C \cap \text{Bd } A \neq \emptyset$.

Proof. Recall that: $\text{Bd } A = \overline{A} \cap \overline{X \setminus A}$.

Suppose that $C \cap \text{Bd } A = \emptyset$. Since $C \cap A \neq \emptyset$, $C \cap A \subseteq C \cap \overline{A} \neq \emptyset$. Similarly, $C \cap \overline{X \setminus A} \neq \emptyset$.

Now, $\overline{A} \cap \overline{X \setminus A} \cap C = \text{Bd } A \cap C = \emptyset$, and $\overline{A} \cup \overline{X \setminus A} \supseteq A \cup X \setminus A \supseteq X \supseteq C$.

Therefore, it is a separation of C . But, C is connected, contradiction.

#9. Let $A \subset X$ be a proper subset, and $B \subset Y$ be a proper subset.

If X and Y are connected, then $(X \times Y) \setminus (A \times B)$ is connected.

Proof. Recall: $X \times Y \setminus A \times B = [(X \setminus A) \times Y] \cup [X \times (Y \setminus B)]$

$(x, y) \in (X \times Y) \setminus (A \times B) \Leftrightarrow (x, y) \notin A \times B \Leftrightarrow x \notin A \text{ or } y \notin B \Leftrightarrow (x, y) \in (X \setminus A) \times Y \text{ or } (x, y) \in X \times (Y \setminus B)$.

Let $a \in X \setminus A$ be given, and let $b \in Y \setminus B$ be given. Clearly, $\{a\} \times Y \cong Y$, $X \times \{b\} \cong X$. So, both are connected, and $(\{a\} \times Y) \cap (X \times \{b\})$ contains (a, b) , so they intersect, so $(\{a\} \times Y) \cup (X \times \{b\})$ is connected. Now,

$$\bigcup_{\substack{a \in X \setminus A \\ b \in Y \setminus B}} (\{a\} \times Y) \cup (X \times \{b\}) = (X \times Y) \setminus (A \times B)$$

is also connected by the problem 3.

#11. Let $p: X \rightarrow Y$ be a quotient map.

If $p^{-1}\{y\}$ is connected for all $y \in Y$, and if Y is connected, then X is connected.

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}\{y\}$$

Suppose that X is disconnected by the separation $\{U, V\}$, that is, opens U, V satisfies $U \neq \emptyset$, $V \neq \emptyset$, $U \cap V = \emptyset$, $U \cup V = X$.

Given $y \in Y$, $p^{-1}\{y\} \subseteq X = U \cup V$. Since $p^{-1}\{y\}$ is connected, either

$$p^{-1}\{y\} \subseteq U \text{ or } p^{-1}\{y\} \subseteq V,$$

because if $p^{-1}\{y\}$ intersects to both U and V , then $\{U, V\}$ becomes separation, for $p^{-1}\{y\}$, but it is contradiction. Observe that:

$$Y = \underset{\text{onto}}{p}[X] = p[U \cup V] = p[U] \cup p[V]$$

Let $a \in p[U]$ be given. That is, $a = p(b) \in p[U]$ for some $b \in U$.

Clearly, $p^{-1}\{a\} \cap U \neq \emptyset$, contains b , so $p^{-1}\{a\} \subseteq U$.

That is, $p^{-1}[p[U]] \subseteq U$, so $p^{-1}[p[U]] = U$, because the reverse inclusion is generally true. Now, $p^{-1}[p[U]]$ is open, so $p[U]$ is open.

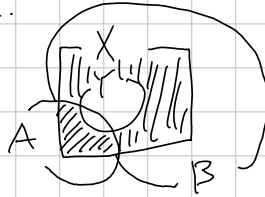
Similarly $p[V]$ is open. So, $p[U]$ and $p[V]$ is the separation for Y , it is contradiction.

#12. Let $Y \neq X$, and both are connected.

If $\{A, B\}$ is a separation of $X \setminus Y$, then $Y \cup A$ and $Y \cup B$ are connected.

Proof. By definition, $A \cap (X \setminus Y) \neq \emptyset$, $B \cap (X \setminus Y) \neq \emptyset$, $A \cap B \cap (X \setminus Y) = \emptyset$, $X \setminus Y \subseteq A \cup B$.

A and B both are open set of X .



Connected of the Real Line.

#1.

a). No two spaces of $(0,1)$, $[0,1]$, $[0,1]$ are Homeomorphic.

Proof If there is a Homeomorphism $f: (0,1] \rightarrow (0,1)$,

then $(0,1] \setminus \{1\} = (0,1) \cong (0,1) \setminus \{f(1)\}$

But, since $f(1) \in (0,1)$, $(0,1) \setminus \{f(1)\}$ is disconnected, by $(0, f(1))$, $(f(1), 1)$.

The Connectedness is preserved by Homeomorphism, it is Contradiction.

Similarly, there is No Homeomorphism $f: [0,1] \rightarrow (0,1)$

Now, Suppose that there is a Homeomorphism $f: [0,1] \rightarrow [0,1]$.

Then, $[0,1] \setminus \{0,1\} = (0,1) \cong (0,1) \setminus f[\{0,1\}]$

But, $(0,1) \setminus \{f(0), f(1)\}$ must be disconnected, so Contradiction.

b). $X = (0,1)$, $Y = [0,1]$.

$f: X \rightarrow Y: x \mapsto x$ is embedding, $g: Y \rightarrow X: x \mapsto \frac{1}{2}x + \frac{1}{4}$ is embedding.

But $X \not\cong Y$.

c).

#2. Let $f: S^1 \rightarrow \mathbb{R}$ be a continuous map. Then, there is a point $x \in S^1$ s.t. $f(x) = f(-x)$.
Proof. Let $t \in S^1$ be given. If $f(t) = f(-t)$, then proved.
Assume $f(t) \neq f(-t)$. Then, either $f(t) - f(-t) > 0$ or < 0 . WLOG, let $f(t) - f(-t) > 0$.
Then, $f(-t) - f(t) < 0$, so, there is a $x \in S^1$ s.t. $f(x) - f(-x) = 0$. ~~pro~~

#9. Assume that A is a countable subset of $\mathbb{R}^2$, then $\mathbb{R}^2 - A$ is path-connected.
Proof. Let (x_1, y_1) and (x_2, y_2) be given from $\mathbb{R}^2 - A$.

Compact space.

1. a). Let $\mathcal{C}, \mathcal{C}'$ be topologies on X , with $\mathcal{C} \subseteq \mathcal{C}'$.

If $(X, \mathcal{C}')$ is compact, then $(X, \mathcal{C})$ is compact.

Proof. Let $\mathcal{O} \subseteq \mathcal{C}$ be an open cover of X . Then, since $\mathcal{O} \subseteq \mathcal{C}'$ and $(X, \mathcal{C}')$ compact, $\mathcal{O}$ has finite subcover. $\square$

1. b). If X is Compact T_2 under both $\mathcal{C}$ and $\mathcal{C}'$,

then either $\mathcal{C}$ and $\mathcal{C}'$ are equal or they are Not comparable.

Proof. If $\mathcal{C} = \mathcal{C}'$, then there is nothing to prove. Suppose $\mathcal{C} \neq \mathcal{C}'$.

Enough to show that neither $\mathcal{C} \subseteq \mathcal{C}'$ nor $\mathcal{C}' \subseteq \mathcal{C}$.

Suppose that $\mathcal{C} \subseteq \mathcal{C}'$. Then, we can choose an open $U \in \mathcal{C}' \setminus \mathcal{C}$.

Now, $X \setminus U$ is closed in $\mathcal{C}'$ but not $\mathcal{C}$. Since $\mathcal{C}'$ is Compact T_2 ,

$X \setminus U$ is Compact set of $\mathcal{C}'$. Since $\mathcal{C} \subseteq \mathcal{C}'$, for any open cover $\mathcal{O} \subseteq \mathcal{C}$ for $X \setminus U$,

it has finite subcover. Therefore, $X \setminus U$ is compact in $(X, \mathcal{C})$, thus closed.

But, it is contradiction. Similarly, $\mathcal{C}' \not\subseteq \mathcal{C}$.

2. a). Every subspace in Co-finite Topology $(\mathbb{R}, \mathcal{C}_f)$ is Compact.

b). In Co-Countable $(\mathbb{R}, \mathcal{C}_c)$, is $[0, 1]$ Compact? an

Proof. Let $A \subseteq \mathbb{R}$ be a subset. Let $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$ be open cover for A . There is,

$$A \subseteq \bigcup_{U \in \mathcal{O}} U \Leftrightarrow \bigcap_{U \in \mathcal{O}} \mathbb{R} \setminus U \subseteq A^c$$

But since $\mathbb{R} \setminus U$ is finite for each $U \in \mathcal{O}$, there are finite number of $U_i, \dots, U_n$ s.t.

$$\bigcap_{i=1}^n \mathbb{R} \setminus U_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^n U_i$$

Therefore A is Compact. (More specifically, for any distinct finite set F_1 and F_2 , $F_1 \cap F_2$ has number of less than $\min\{\#(F_1), \#(F_2)\}$.)

But, in the Co-Countable, $[0, 1]$ is Not compact, because

Construct sequences; denote $\mathcal{O} \cap [0, 1] = \{U_n \mid n \in \mathbb{N}\}$, and let $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$

Then, $\bigcup_{n=1}^{\infty} U_n \supseteq [0, 1]$ but it has no finite subcover.

3. Finite Union of Compact Subspace is Compact.

Proof. Let $K_1, \dots, K_n \subseteq X$ be Compact subsets. Let $\mathcal{O} \subseteq \mathcal{P}(X)$ be an open cover of $\bigcup_{i=1}^n K_i$.

Since $\bigcup_{U \in \mathcal{O}} U \supseteq \bigcup_{i=1}^n K_i \supseteq K_j$ ($1 \leq j \leq n$), for each j , $\mathcal{O}$ has a finite subcover for K_j ,

denote $\mathcal{U}_j^1, \dots, \mathcal{U}_j^{r_j} \in \mathcal{O}$. Now, clearly

$$\bigcup_{i=1}^n K_i \subseteq \bigcup_{j=1}^n \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

4. Every Compact Subspace in a Metric space is closed bounded.

Give the example that the Converse fails.

Proof. Closed is given by: Compact in T_2 is closed, because

Let X be a Hausdorff space, and $K \subseteq X$ be a Compact subset.

Let $x \in X \setminus K$ be fixed. Now, for each $y \in K$, there exists disjoint open $\mathcal{U}_y^x, \mathcal{U}_y$ s.t.

$$x \in \mathcal{U}_y^x, y \in \mathcal{U}_y, \mathcal{U}_y^x \cap \mathcal{U}_y = \emptyset$$

Now, $\{\mathcal{U}_y \mid y \in K\}$ is open cover of K , therefore there is a finite subcover

$\{\mathcal{U}_{y_i} \mid i=1, 2, \dots, r_x\}$ and open neighborhoods of x $\{\mathcal{U}_{y_i}^x \mid i=1, \dots, r_x\}$.

Put $V_x = \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}^x$. Then, V_x is open neighborhood and $K \cap V_x = \emptyset$, thus $V_x \subseteq K^c$.

(More specifically, $K \subseteq \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}$ and $\left(\bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}\right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{U}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$).

And boundedness is clear.

Counterexample: Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, discrete topology.

Then, it is closed itself, and bounded, but not Compact.

15. Let X be a Hausdorff, and $A, B \subseteq X$ be disjoint compact subsets.

Then, there exist disjoint open sets U and V containing A and B , respectively.

Proof. Let $x \in A$ be fixed. Then, for each $y \in B$, there exist opens $U_{(x,y)}$ and $V_{(x,y)}$ s.t.

$$x \in U_{(x,y)}, \quad y \in V_{(x,y)}, \quad U_{(x,y)} \cap V_{(x,y)} = \emptyset.$$

Since $\{V_{(x,y)} \mid y \in B\}$ is open cover of B , there exists a finite subcover

$$\{V_{(x,y_i)} \mid i=1, 2, \dots, r_x\}. \text{ Now, } U_x = \bigcap_{i=1}^{r_x} U_{(x,y_i)} \text{ is open neighborhood of } x.$$

$$V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)} \text{ is open set containing } B, \text{ and } U_x \cap V_x = \emptyset. \text{ because}$$

$$U_x \cap V_x = U_x \cap \left(\bigcup_{i=1}^{r_x} V_{(x,y_i)} \right) = \bigcup_{i=1}^{r_x} (U_x \cap V_{(x,y_i)}) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$$

In summary, for each $x \in A$, there are two disjoint opens U_x, V_x s.t.

$$x \in U_x, \quad B \subseteq V_x, \quad U_x \cap V_x = \emptyset.$$

And, $\{U_x \mid x \in A\}$ is an open subcover of A , thus it has a finite subcover

$$\{U_{x_i} \mid i=1, \dots, m\}. \text{ Finally,}$$

$$\bigcup_{i=1}^m U_{x_i} \text{ and } \bigcap_{i=1}^m V_{x_i} \text{ are disjoint open sets containing } A \text{ and } B, \text{ respectively.}$$

$$\text{Disjointness is given by } \left(\bigcup_{i=1}^m U_{x_i} \right) \cap \left(\bigcap_{i=1}^m V_{x_i} \right) = \bigcup_{i=1}^m \left(U_{x_i} \cap \left(\bigcap_{i=1}^m V_{x_i} \right) \right) = \bigcup \emptyset = \emptyset.$$

16. If $f: X \rightarrow Y$ is continuous map where X is compact and Y is Hausdorff, then f is closed map.

Proof. Let $C \subseteq X$ be a closed set. Since X is compact, C is compact.

And, the continuous map f preserves compactness, $f[C]$ is compact in T_2 , thus closed.

17. If Y is compact, then the projection $p_X: X \times Y \rightarrow X$ is a closed map.

Proof. Let $C \subseteq X \times Y$ be a closed set. Want to show that $X \setminus p_X[C]$ is open.

Let $x \in X \setminus p_X[C]$ be given. Then, $\{x\} \times Y \cap C = \emptyset$, therefore $\{x\} \times Y \subseteq (X \times Y) \setminus C$.

Now, by the Tube Lemma there is an open $U \subseteq X$ s.t.

$$\{x\} \times Y \subseteq U \times Y \subseteq (X \times Y) \setminus C.$$

Now, $x \in U$ and $U \subseteq X \setminus p_X[C]$, thus $X \setminus p_X[C]$ is open.

Note: the converse is true, but so hard to prove.

Recall: The Tube Lemma.

Let X be a topological space, and Y be a compact space.

Then, for the product space $X \times Y$, for fixed $x_0 \in X$,

For any open $N \subseteq X \times Y$ s.t. $\{x_0\} \times Y \subseteq N$, there is open $W \in \mathcal{C}_X$ s.t. $\{x_0\} \times Y \subseteq W \times Y \subseteq N$.

Proof. Since $\{x_0\} \times Y \rightarrow Y: (x_0, y) \mapsto y$ is homeo, $\{x_0\} \times Y$ is compact.

Since N is open, for any $y \in Y$, $\exists U_y \in \mathcal{C}_X, V_y \in \mathcal{C}_Y$ s.t. $\{x_0\} \times y \subseteq U_y \times V_y \subseteq N$.

Now, open cover $\{U_y \times V_y \mid y \in Y\}$ of $\{x_0\} \times Y$ has a finite subcover

$$\{U_{y_i} \times V_{y_i} \mid i=1, \dots, N\}.$$

Set $W = \bigcap_{i=1}^N U_{y_i}$. Then, $\{x_0\} \times Y \subseteq W \times Y$ is clear, and

$$(x, y) \in W \times Y \Leftrightarrow \forall i=1, \dots, N, x \in U_{y_i} \text{ and } y \in V_{y_i}$$

$$\Rightarrow (x, y) \in U_{y_1} \times V_{y_1} \subseteq N.$$

8. Let X be any space, Y be a Compact Hausdorff. Then,

$f: X \rightarrow Y$ is continuous $\xleftrightarrow{T_2}$ if and only if $G_f = \{(x, f(x)) \mid x \in X\} \subseteq X \times Y$ is closed.

Proof. Suppose that f is continuous. Let $(x, y) \in (X \times Y) \setminus G_f$ be given. Clearly $y \neq f(x)$. Since Y is T_2 , there exist disjoint opens $U, V \subseteq Y$ containing $y, f(x)$, respectively.

In other words, $(y, f(x)) \in U \times V \subseteq (Y \times Y) \setminus \Delta(Y)$

And $f(x) \in V$ implies $x \in f^{-1}[f[x]] \subseteq f^{-1}[V] \subseteq X$. Now enough to show that only (x, y)
 $(x, y) \in f^{-1}[V] \times U \subseteq (X \times Y) \setminus G_f$

because f is continuous, thus $f^{-1}[V]$ is open. The inclusion (*) equals to:

$$(f^{-1}[V] \times U) \cap G_f = \emptyset$$

and it is true, if it is not empty set, then for some $(x, f(x)) \in G_f$,
 $(x, f(x)) \in f^{-1}[V] \times U$. That is, $f(x) \in V$ and $f(x) \in U$, which is contradiction.

(Note: in this direction we used only T_2 condition.)

Conversely, suppose that the graph G_f is closed set.

Let $U \subseteq Y$ be an open set. Want to show that: $f^{-1}[U]$ is an open set of X .

Let $x \in f^{-1}[U]$ be given. Then, $f(x) \in U$ by definition.

Let $x_0 \in X$ be given. Then, for any open $V \subseteq Y$ containing $f(x_0)$,

$$G_f \cap (X \times (Y \setminus V))$$

is closed set. Now, since Y is compact, the projection $\pi_X: X \times Y \rightarrow X$ is closed map, thus

$$\pi_X[G_f \cap (X \times (Y \setminus V))] \subseteq X \text{ is closed in } X.$$

Now, $U = X \setminus \pi_X[G_f \cap (X \times (Y \setminus V))]$ is open, moreover, $x_0 \in U$ because

$$x_0 \in U \Leftrightarrow x_0 \notin \pi_X[G_f \cap (X \times (Y \setminus V))] \Leftrightarrow \nexists (x, y) \in G_f \cap (X \times (Y \setminus V)), \pi_X(x, y) = x \neq x_0$$

But, since $(x_0, f(x_0)) \notin (X \times (Y \setminus V))$, the last assertion holds. Now, only showing $f^{-1}[U] \subseteq V$ is remained.

Let $x \in U$. Then, $f(x) \notin Y \setminus V$, that is, $f(x) \in V$. ~~is~~

8. Revised. Let X be any space and Y be Compact Hausdorff space.

$f: X \rightarrow Y$ is continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

Proof. Let $f: X \rightarrow Y$ be a continuous map.

Let $(x, y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets U, V

$$U \cap V = \emptyset, \quad y \in U, \quad f(x) \in V$$

Since $f(x) \in V \stackrel{\text{def}}{\Leftrightarrow} x \in f^{-1}[V]$ and f is continuous, $f^{-1}[V]$ is an open set in X .

Now, $(x, y) \in f^{-1}[V] \times U$

And, $(f^{-1}[V] \times U) \cap G_f = \emptyset$, because if $(x, f(x)) \in f^{-1}[V] \times U$, then

$f(x) \in V$ and $f(x) \in U$, it is contradiction to disjointness.

Therefore, $(x, y) \in f^{-1}[V] \times U \subseteq X \times Y \setminus G_f$, so proved. $\Rightarrow$

Conversely, let G_f is closed in $X \times Y$, under the Product Topology.

Let $C \subseteq Y$ be a closed set in Y . Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$p_x[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $p_x: X \times Y \rightarrow X: (x, y) \mapsto x$ is closed map if Y is compact. $\square$

Note: for $g = f \times i: X \times Y \rightarrow Y \times Y: (x, y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x, y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f \quad \square \Leftarrow$$

Product of Compact space is compact.

Let X, Y be a Compact space.

Let $\mathcal{O}$ be an open cover for $X \times Y$. Given $x_0 \in X$, $\{x_0\} \times Y \subseteq Y$,

So there is finite subcover

$$\{x_0\} \times Y \subseteq \bigcup_{i=1}^{N_{x_0}} U_{x_0}^i \quad (U_{x_0}^i \in \mathcal{O})$$

Now, by the tube lemma $\exists$ open $N_{x_0} \subseteq X$ s.t.

$$\{x_0\} \times Y \subseteq N_{x_0} \times Y \subseteq \bigcup_{i=1}^{N_{x_0}} U_{x_0}^i.$$

$$(\cup A) \times Y$$

$$= \cup (A \times Y)$$

79. The Generalized Tube Lemma.

Let $A \subseteq X$, $B \subseteq Y$ be compact subspaces. Then,

For any open $N \subseteq X \times Y$ containing $A \times B$, there exists opens $U \subseteq X$ and $V \subseteq Y$ s.t. $A \times B \subseteq U \times V \subseteq N$.

Proof. Let $N \subseteq X \times Y$ be the open set such that

$$A \times B \subseteq N \subseteq X \times Y.$$

Given $(x, y) \in N$, there exists open $U_x \subseteq X$ and $U_y \subseteq Y$ such that

$$(x, y) \in U_x \times U_y \subseteq N.$$

Clearly, $\{U_x \times U_y \mid (x, y) \in N\}$ is open covering of N , so it covers $A \times B$.

12. Let $p: X \rightarrow Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is Compact.

If Y is Compact, then X is Compact. (Such p is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let $\mathcal{O} = \{U_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X . For each $y \in Y$,

$$p^{-1}[\{y\}] \subseteq \bigcup_{y \in Y} p^{-1}[\{y\}] \subseteq \bigcup_{\alpha \in \Lambda} U_\alpha$$

And by the assumption, $p^{-1}[\{y\}]$ is Compact, therefore there is a finite index $\Lambda_y \subseteq \Lambda$ s.t.

$$p^{-1}[\{y\}] \subseteq \bigcup_{i \in \Lambda_y} U_i.$$

Now, since p is closed map,

$$V_y = Y \setminus p[X \setminus (\bigcup_{i \in \Lambda_y} U_i)]$$

is an open set of Y , containing y . Now, since Y is compact, for some $y_1, \dots, y_n$,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^n V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\left[\bigcup_{j=1}^n V_{y_j}\right] = \bigcup_{j=1}^n p^{-1}[V_{y_j}]$$

Meanwhile, $p^{-1}[V_y] = p^{-1}\left[Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} U_i\right)\right]\right] = X \setminus p^{-1}\left[p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} U_i\right)\right]\right] \subseteq \bigcup_{i \in \Lambda_y} U_i,$

therefore $Y \subseteq \bigcup_{j=1}^n \bigcup_{i \in \Lambda_{y_j}} U_i$

Γ + Extra problem

#1. Let X be a Hausdorff space, and $K_\alpha \subseteq X$ ($\alpha \in \Lambda$) be compact subsets. If $\{K_\alpha \mid \alpha \in \Lambda\}$ satisfies F.I.P., then $\bigcap_{\alpha \in \Lambda} K_\alpha \neq \emptyset$.

Proof. Suppose that $\bigcap_{\alpha \in \Lambda} K_\alpha = \emptyset$. Let $\beta \in \Lambda$ be fixed. Then,

$$K_\beta \cap \left(\bigcap_{\alpha \neq \beta} K_\alpha \right) = \bigcap_{\alpha \in \Lambda} K_\alpha = \emptyset, \text{ thus } K_\beta \subseteq \bigcup_{\alpha \neq \beta} K_\alpha^c$$

Since X is T_2 , K_α^c is open, so $\{K_\alpha^c \mid \alpha \in \Lambda \setminus \{\beta\}\}$ is open cover of K_β . Now, there is a finite subcover so

$$K_\beta \subseteq \bigcup_{i=1}^n K_{\alpha_i}^c, \text{ that is, } K_\beta \cap \left(\bigcap_{i=1}^n K_{\alpha_i} \right) = \emptyset.$$

It is contradiction to F.I.P.

#2. Let X be a compact set, and $A \subseteq X$ be an infinite subset. Then, $A' \neq \emptyset$.

Proof. Suppose that $A' = \emptyset$. Then, for any $x \in X$, there is an open U_x containing x

$$A \cap U_x \setminus \{x\} = \emptyset.$$

Since $\{U_x \mid x \in X\}$ is open cover of X , there is a finite subcover

$$\{U_{x_i} \mid i=1, 2, \dots, n\}.$$

Then, $A \subseteq X = \bigcup_{i=1}^n U_{x_i}$, but $A = A \cap \left(\bigcup_{i=1}^n U_{x_i} \right) = \bigcup_{i=1}^n (A \cap U_{x_i}) \subseteq \bigcup_{i=1}^n \{x_i\}$, which is contradiction.

#3. Co-Countable space $(\mathbb{R}, \mathcal{C}_c)$.

In $(\mathbb{R}, \mathcal{C}_c)$, $A \subseteq \mathbb{R}$ is Compact $\Leftrightarrow A$ is finite.

Proof. If A is finite, then trivial. If A is infinite, construct $\{x_n\}$ as

$x_1 \in A$, $x_2 \in A \setminus \{x_1\}$, $x_3 \in A \setminus \{x_1, x_2\}$, ..., then,

$U_n = \mathbb{R} \setminus \{x_i \mid i \in \mathbb{N} \setminus \{n\}\}$ is open of the given space, and

$A \subseteq \bigcup_{n \in \mathbb{N}} U_n$ but, it has no finite subcover

If $f: \mathbb{R} \rightarrow (\mathbb{R}, \mathcal{C}_c)$ is continuous, it is Contant map.

Proof. Since the Compactness is preserved by Continuous, for any $[a, b]$,

$f[[a, b]]$ is compact in $(\mathbb{R}, \mathcal{C}_c)$.

That is, $f[[a, b]]$ is finite for any $a < b$. Now,

$f[\mathbb{R}] = f\left[\bigcup_{n \in \mathbb{Z}} [n, n+1]\right] = \bigcup_{n \in \mathbb{Z}} f[[n, n+1]]$ is at most Countable.

Meanwhile, since the Closedness is preserved by Continuous Preimage,

every finite or countable subset C , $f^{-1}[C]$ is closed.

Denote $\text{Im } f = \{x_n \mid n \in \mathbb{N}\}$, Let $A = f^{-1}[\{x_1\}]$, $B = f^{-1}[\{x_n \mid n \in \mathbb{N} \setminus \{1\}\}]$.

Then, $\mathbb{R} = A \cup B$ and $A \cap B = \emptyset$. And both A and B are closed,

$A = \mathbb{R} \setminus B$ is clopen, $A = \mathbb{R}$. Contradiction.

* Fail Try of 12.

T2. Let $p: X \rightarrow Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Let $\mathcal{C} \subseteq \mathcal{C}_X$ be an open cover of X . That is, $\bigcup_{U \in \mathcal{C}} U = X$
Now, by the surjectivity, $p\left[\bigcup_{U \in \mathcal{C}} U\right] = \bigcup_{U \in \mathcal{C}} p[U] = Y$.

And, since p is closed onto map, it is open map. Therefore, $p[U]$ ($U \in \mathcal{C}$) is open cover of Y , thus it has finite subcover s.t. $\bigcup_{i=1}^N p[U_i] = Y$, since Y is compact. Observe that

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{i=1}^N p[U_i]\right] = \bigcup_{i=1}^N p^{-1}[p[U_i]]$$

Let $y \in Y$ be given. By the surjectivity, $p^{-1}[\{y\}]$ is non-empty.

Let $x_0 \in X$ s.t. $p(x_0) = y$.

$$\begin{aligned} p^{-1}[W] &\subseteq U \\ \Leftrightarrow W &\subseteq p[U] \end{aligned}$$

Let $y \in Y$ be given. If $U \subseteq X$ is open containing $p^{-1}[\{y\}]$, then

$$p^{-1}[\{y\}] \subseteq U \Rightarrow p[p^{-1}[\{y\}]] \subseteq p[U] \Rightarrow y \in p[U]$$

$$\Rightarrow y \in W \subseteq p[U]$$

$$Y = \bigcup_{y \in Y} (Y \setminus p[X \setminus (\bigcup_{i \in \mathbb{N}} U_i)]) = \bigcup_{j=1}^n (Y \setminus p[X \setminus (\bigcup_{i \in \mathbb{N}_j} U_i)])$$

Ex. 27.

2. Let (X, d) be a metric space, and let $A \subset X$ be non-empty.

a). $d(x, A) = 0 \Leftrightarrow x \in \bar{A}$

b). If A is compact, then $d(x, A) = d(x, a)$ for some $a \in A$

c). $\mathcal{U}(A, \epsilon) \stackrel{\text{def}}{=} \{x \in X \mid d(x, A) < \epsilon\} = \bigcup_{a \in A} B_\epsilon(a)$, for any $\epsilon > 0$.

d). If A is compact, and $\mathcal{U}$ be an open containing A .

Then for some $\epsilon > 0$,

$$A \subseteq \mathcal{U}(A, \epsilon) \subseteq \mathcal{U}.$$

e). Find a) fails.

Proof. For $x \in X$,

$$d(x, A) = 0 \Leftrightarrow \inf\{d(x, a) \mid a \in A\} = 0$$

$$\Leftrightarrow \text{given } \epsilon > 0 \text{ there is } a \in A \text{ s.t. } 0 \leq d(x, a) < \epsilon$$

$$\Leftrightarrow \quad \quad \quad \text{s.t. } a \in B_\epsilon(x)$$

$$\Leftrightarrow x \in \bar{A}.$$

And, for fixed $x \in X$, $f(a) = d(x, a)$ is compact, so $f[A]$ is compact.

Since $d(x, A) = \inf f[A]$, by the min-max theorem, it has a minimum in A .

Let $x \in \mathcal{U}(A, \epsilon)$. Then, $d(x, A) < \epsilon$. Since $d(x, A) = \inf\{d(x, a) \mid a \in A\} < \epsilon$, for some $a \in A$, $d(x, a) < \epsilon$. (if $d(x, a) \geq \epsilon$ for all a , then $d(x, A) \geq \epsilon$) so, $x \in B_\epsilon(a)$. The converse is trivial.

Ex. Connected Metric space having more than one point is uncountable.

Proof. Let $a, b \in X$ be distinct points in X .

Let $\lambda \in (0, 1)$ be given. If there is no such a point $x \in X$ s.t. $d(a, x) \neq \lambda \cdot d(a, b)$, then $U = \{z \in X \mid d(a, z) < \lambda \cdot d(a, b)\}$ and $V = \{z \in X \mid d(a, z) > \lambda \cdot d(a, b)\}$

are disjoint non-empty open sets, cover the all X .

Therefore for each $\lambda \in (0, 1)$, there is x_λ s.t.

$$d(a, x_\lambda) = \lambda \cdot d(a, b),$$

so $(0, 1) \rightarrow X: \lambda \mapsto x_\lambda$ is injective, X is uncountable.

Thm. Let X be a non-empty Compact Hausdorff space.

If X has no isolated points, then X is uncountable.

Proof. Let $U \subseteq X$ be a non-empty open set.

Choose $x \in U$. And, choose $y \in U$, with $y \neq x$.

(If no such y exists, then $U = \{x\}$, then x is the isolated point, it is contradiction.)

And, since X is Hausdorff, there are disjoint open sets W_1, W_2 such that

$$x \in W_1 \text{ and } y \in W_2.$$

Now, let $V = \overline{W_2 \cap U}$. Then, it is non-empty (containing y) and

$x \notin V = \overline{W_2 \cap U}$, because $x \in \overline{W_2 \cap U} \subseteq \overline{W_2 \cap U}$, then $x \in \overline{W_2}$. It means $W_1 \cap W_2 \neq \emptyset$.

Now, let $x_1 \in X$. Then, there is $x_2 \neq x_1$ in X and there is an open $V_1 \subseteq X$ s.t. $x_1 \notin \overline{V_1}$ but $x_2 \in V_1$. For this x_2 , again, choose x_3 and V_2 s.t.

$$V_2 \text{ open, } V_2 \subseteq V_1, x_3 \in V_2, x_2 \notin \overline{V_2}.$$

Inductively, $\overline{V_1} \supseteq \overline{V_2} \supseteq \dots$ and $\overline{V_n}$ is nonempty for all n .

So, there is a point $x \in \bigcap \overline{V_n}$, but $x \neq x_n$ for all n , by the construction.

That is, $f: \mathbb{N} \rightarrow X: n \mapsto x_n$ is injective but not surjective, so X is uncountable.

Def. Let X be a space. We said X is locally compact at $x \in X$ if:

there exists open U and compact C such that $x \in U \subseteq C$.

Thm. Let X be a Hausdorff space. Then, X is locally compact if and only if

for any $x \in X$, there exists an open U containing x such that $\overline{U}$ is compact.

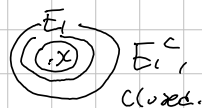
Proof. Let X be a locally-compact T_2 . Given $x \in X$, there are open U and compact C s.t.

$$x \in U \subseteq C$$

Since compact in T_2 is closed, so $\overline{U} \subseteq C$. And closed in compact is compact, this $\overline{U}$ is the compact set we desired.

Note: Clearly, Compact space is locally compact.

Ex.



Let X be a Compact Hausdorff space. So, it is Normal space. Given $U \subseteq X$ be open. Let $\{A_n\}$ be a Countable dense open sets of X . Then, $\bigcap_{n \in \mathbb{N}} A_n$ is dense in X .

Proof. Let $E_1 = U \cap A_1$. Since A_1 is dense in X , E_1 is non-empty, so, let $x_1 \in E_1$. And, since U and A_1 are open, so E_1 is open, so there an open E_2 such that

$$x_1 \in E_2 \subseteq \overline{E_2} \subseteq \overline{E_1}$$

because it is normal, so regular. Now, let $x_2 \in \overline{E_2} \cap A_2$. Similarly, Construct $\{E_n\}$:

$$\overline{E_n} \text{ non-empty-compact} \quad \overline{E_{n+1}} \subseteq \overline{E_n} \cap A_{n+1} \text{ for all } n \in \mathbb{N}.$$

Since $\overline{E_n}$ are compact, $\bigcap_{n \in \mathbb{N}} \overline{E_n}$ is not empty because: if $\bigcap_{n \in \mathbb{N}} \overline{E_n} = \emptyset$, then

$$\bigcup_{n \in \mathbb{N}} \overline{E_n}^c = X, \text{ so } \bigcup_{n \in F} \overline{E_n}^c \text{ for some finite } F \subseteq \mathbb{N}, \text{ since } X \text{ is compact.}$$

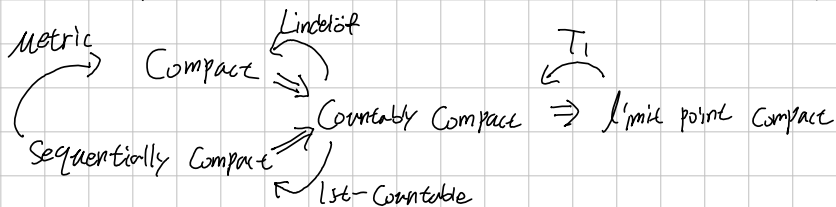
But, $\{\overline{E_n}\}$ is nested non-empty sets, the finite intersection cannot be empty.

Therefore, $\emptyset \neq \bigcap_{n \in \mathbb{N}} \overline{E_n} \subseteq (\bigcap_{n \in \mathbb{N}} A_n) \cap U$, so proved.

Section 28, Limit Point Compactness.

Def. A space X is said to be:

- 1) Limit point Compact if every infinite subset of X has a limit point.
- 2) Sequentially Compact if every sequence has a convergent subsequence.



Thm. Countably Compact $\Rightarrow$ Limit Point Compact.

Proof. Let $A \subseteq X$ be an infinite subset of Countably Compact set X .

Since A is infinite, there exists a countable infinite subset $\{x_n \in A \mid n \in \mathbb{N}\} \subseteq A$. Suppose that $\{x_n \in A \mid n \in \mathbb{N}\}$ has no limit point.

Then, for each $k \in \mathbb{N}$, $C_k = \{x_n \mid n \geq k\}$ are closed, since $C_k^c = \emptyset$.

Moreover, $\bigcap_{k=1}^{\infty} C_k = \emptyset$, or $\bigcup_{k=1}^{\infty} C_k^c = X$. But, since X is Countably Compact,

For some finite subset $F \subseteq \mathbb{N}$, $\bigcup_{k \in F} C_k^c = X$, or $\bigcap_{k \in F} C_k = \emptyset$.

But, $x_{\max F + 1}$ must be contained in $\bigcap_{k \in F} C_k$, thus it is contradiction.

Thm. Sequentially Compact $\Rightarrow$ Countably Compact.

Proof. Using Contraposition. Let X be a not Countably compact space.

Then, there exists a countable open cover $\mathcal{O} = \{U_n \mid n \in \mathbb{N}\}$ which has no finite subcover.

Thus, for each $n \in \mathbb{N}$, $X \setminus \bigcup_{i=1}^n U_i \neq \emptyset$, let $x_n \in X \setminus \bigcup_{i=1}^n U_i$ for each $n \in \mathbb{N}$.

Let $\{x_{n_k}\}$ be an arbitrary subsequence for $\{x_n\}$.

For any $x \in X$, since $\mathcal{O}$ is open cover for X , for some $j \in \mathbb{N}$, $x \in U_j$.

But for all $k \geq j$, $n_k \geq k$, so $x_{n_k} \notin U_j$. That is, $\{x_{n_k}\}$ does not converge any point in X .

Sec. Countability.

#2. Let X be the second Countable.

Then, every basis $\mathcal{C}$ for X contains a Countable basis.

Proof. Let $\mathcal{B} = \{B_n \mid n \in \mathbb{N}\}$ be a Countable basis.

Let $\mathcal{C}' = \{C_{m,n} \in \mathcal{C} \mid m, n \in \mathbb{N}, B_n \subseteq C_{m,n} \subseteq B_m\} \cup \{\emptyset\}$.

Then $\mathcal{C}'$ is Countable subset of $\mathcal{C}$. And, for any open $U \subseteq X$ and $x \in U$, there is $B_m \in \mathcal{B}$ s.t. $x \in B_m \subseteq U$, and $C \in \mathcal{C}$ s.t. $x \in C \subseteq B_m$.

Now, for some B_n , $x \in B_n \subseteq C \subseteq B_m \subseteq U$, $C \in \mathcal{C}'$, so proved.

#3. Uncountable subset of second countable set has uncountable limit point in itself.

Proof. Let $\mathcal{B}$ be the Countable base and A be the uncountable subset.

If $A \cap A'$ is Countable, denote $A \cap A' = \{x_n \mid n \in \mathbb{N}\}$.

For each x_n and basis $B_m \in \mathcal{B}$, $(B_m \setminus \{x_n\}) \cap A \neq \emptyset$

#4. Compact Metric space is 2nd-Countable.

Proof. Let $\mathcal{B}_n = \{B_{\frac{1}{n}}(x) \mid x \in X\}$, ($n \in \mathbb{N}$) Then, it is open cover for $\mathcal{B}$.

So there is finite subcover $\mathcal{B}'_n = \{B_{\frac{1}{n}}(x_i) \mid i=1, 2, \dots, r_n\}$.

Now, let $\mathcal{B}' = \bigcup_{n \in \mathbb{N}} \mathcal{B}'_n$. Then, it is Countable union of finite collections, so Countable.

Let $x \in X$ and open U containing x be given. Then, for some $N \in \mathbb{N}$,

$$x \in B_{\frac{1}{N}}(x) \subseteq U.$$

Now, since $\mathcal{B}'_{2N}$ covers X , for some $i=1, \dots, r_{2N}$, $x \in B_{\frac{1}{2N}}(x_i) \subseteq B_{\frac{1}{N}}(x) \subseteq U$.

so proved.